

NMEC595 - Research Methodology

Lecture 10: Probability Distribution

(Reference: Research Methodology, C.R. Kothari, 2nd Edition)

Prof. Suparno Bhattacharyya

suparno@iitism.ac.in

Room 224A, Mechanical Engg. Building

Department of Mechanical Engineering
Indian Institute of Technology (ISM) Dhanbad

Winter Semester 2026

Outline

- 1 Introduction to Random Variables
- 2 Probability Distribution Functions
- 3 Discrete Probability Distribution Distributions
- 4 Binomial Distribution
- 5 Continuous Distributions
- 6 Normal Distribution
- 7 Summary and Key Takeaways

Learning Objectives

Upon completion of this module, you should be able to:

- 1 Distinguish between discrete and continuous random variables
- 2 Compute probabilities, cumulative probabilities, means, and variances
- 3 Identify and work with binomial random variables
- 4 Calculate probabilities of binomial random variables
- 5 Describe properties of the normal distribution
- 6 Find probabilities and percentiles of normal distributions
- 7 Apply the Empirical Rule

Motivation: Why Random Variables?

Scenario: Flip a fair coin 3 times

Sample Space:

$$S = \{HHH, HHT, HTH, THH, TTT, TTH, THT, HTT\}$$

- 8 possible outcomes
- For 4 flips: $2^4 = 16$ outcomes
- For 10 flips: $2^{10} = 1024$ outcomes!

Solution: Instead of tracking all outcomes, define X = number of heads
For $n = 3$: X can be 0, 1, 2, or 3

Definition: Random Variable

Random Variable

A **random variable** is a variable that takes on different values determined by chance. It is a numerical quantity that varies at random. You can assign a probability to each of the value that the random variable may take.

Notation Convention:

- Capitalized letters (e.g., X , Y) represent random variables
- Lowercase letters (e.g., x , y) represent specific values

Types of Random Variables

Discrete Random Variable

Can assume only a **countable** number of values (sometimes infinite)

Examples:

- Number of defective parts in a batch
- Number of customers arriving per hour
- Number of failures before first success

Continuous Random Variable

Can assume an **uncountable** number of values in an interval

Examples:

- Temperature measurements
- Material strength (tensile, yield)
- Manufacturing dimensions
- Response time of a system

Probability Distribution Functions: Overview

Probability Distribution Function

A **Probability Distribution function** provides probabilities for possible outcomes of a random variable X . Denoted as $f(x)$.

Two classes:

- ➊ **Probability Distribution Mass Function (PMF)** – for discrete variables
- ➋ **Probability Distribution Density Function (PDF)** – for continuous variables

Why use distributions?

- **Concise summary:** One function $P(X)$ replaces tedious counting
- **Modeling:** PDF models a random process; enables hypothesis testing & parameter estimation
- **Visualize & Compute:** Plot $P(X=k)$; compute $P(4 \leq X \leq 6)$ via $\sum$ or $\int_4^6 f(x) dx$
- **Manipulate functions on X :** e.g., X^2 , propagate through dynamics, Gaussian Processes (GP)

Probability Distribution Mass Function (PMF)

PMF for Discrete Random Variables

If X is discrete: $f(x) = P(X = x)$

The PMF gives the Probability Distribution that X equals x

Properties:

- 1 $f(x) > 0$ for x in the sample space, and 0 otherwise
- 2 $\sum_x f(x) = 1$ (sum of all probabilities equals 1)

Can be expressed as:

- Equation/formula
- Table

Example: PMF in Tabular Form

Data set: $\{0, 1, 2, 3, 4\}$ with equal probabilities

If we randomly draw from these values:

x	0	1	2	3	4
$f(x) = P(X = x)$	1/5	1/5	1/5	1/5	1/5

As a function:

$$f(x) = \begin{cases} \frac{1}{5} & x = 0, 1, 2, 3, 4 \\ 0 & \text{otherwise} \end{cases}$$

Verify: $\sum_{x=0}^4 f(x) = 5 \times \frac{1}{5} = 1 \checkmark$

Probability Density Function (PDF)

Continuous random variables use a *probability density function* (PDF) to describe probabilities.

Definition

For a continuous variable X , the probability that X falls in $[a, b]$ is:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Key properties:

- $f(x) \geq 0$ for all x
- $\int_{-\infty}^{\infty} f(x) dx = 1$ (total area under the curve)
- $P(X = x) = 0$ for any specific x (probabilities are only meaningful over intervals)

Interpretation: $f(x)$ gives the *density* at x , not the probability at x .

Probability Distribution Density Function (PDF)

Note

For PDFs,

$$P(a < Y < b) = P(a \leq Y < b) = P(a < Y \leq b) = P(a \leq Y \leq b)$$

Including or excluding endpoints doesn't matter!

PDF Examples: Uniform and Normal

Uniform Distribution: $X \sim \text{Uniform}(0, 4)$

$$f(x) = \begin{cases} \frac{1}{4} & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Example: $P(1 \leq X \leq 3) = \int_1^3 \frac{1}{4} dx = \frac{1}{2}$

Normal Distribution: $X \sim \mathcal{N}(0, 1)$

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

Example: $P(-1 \leq X \leq 1) = \int_{-1}^1 f(x) dx \approx 0.683$

Note: For both, $P(X = x) = 0$ for any specific x .

PMF vs PDF: Discrete vs Continuous

Discrete random variables use a *probability mass function* (PMF), while **continuous variables** use a PDF.

	PMF (Discrete)	PDF (Continuous)
Definition	$P(X = x)$	Density at x
How to find probability	$P(X = x)$ directly	Integrate: $P(a \leq X \leq b) = \int_a^b f(x) dx$
Probability at a point	Can be > 0	Always 0
Total probability	$\sum_x f(x) = 1$	$\int_{-\infty}^{\infty} f(x) dx = 1$
Examples	Binomial, Poisson	Normal, Uniform, Exponential

Summary: PMFs give probabilities at points; PDFs require integration over intervals.

Cumulative Distribution Function (CDF)

CDF Definition

The **cumulative distribution function** $F(x)$ gives the Probability Distribution that random variable X is less than or equal to value x :

$$F(x) = P(X \leq x)$$

Key Points:

- Same definition for discrete and continuous variables
- For continuous: $P(X = x) = 0$, so $P(X \leq x) = P(X < x)$
- CDF is always non-decreasing
- $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$

Example: Computing CDF

Given PMF:

x	0	1	2	3	4
$f(x) = P(X = x)$	1/5	1/5	1/5	1/5	1/5

Step-by-step CDF calculation:

$$F(0) = P(X \leq 0) = P(X = 0) = \frac{1}{5}$$

$$F(1) = P(X \leq 1) = P(X = 0) + P(X = 1) = \frac{2}{5}$$

$$F(2) = P(X \leq 2) = \frac{1}{5} + \frac{1}{5} + \frac{1}{5} = \frac{3}{5}$$

$$F(3) = P(X \leq 3) = \frac{4}{5}$$

$$F(4) = P(X \leq 4) = \frac{5}{5} = 1$$

Example: CDF Table

Resulting CDF:

x	0	1	2	3	4
$F(x) = P(X \leq x)$	1/5	2/5	3/5	4/5	1

Using the CDF:

- $P(X = 1) = F(1) - F(0) = \frac{2}{5} - \frac{1}{5} = \frac{1}{5}$
- $P(X \leq 2) = F(2) = \frac{3}{5}$
- $P(X < 3) = P(X \leq 2) = \frac{3}{5}$
- $P(1 \leq X \leq 3) = F(3) - F(0) = \frac{4}{5} - \frac{1}{5} = \frac{3}{5}$

CDF for Continuous Variables

For continuous variables:

$$F(y) = P(Y \leq y) = P(Y < y)$$

(since $P(Y = y) = 0$)

Finding probabilities:

- $P(Y > a) = 1 - F(a)$
- $P(a < Y < b) = F(b) - F(a)$

Note: Calculus required to find $F(y)$ from $f(y)$. We use tables/software!

Expected Value (Mean)

Expected Value of Discrete Random Variable

The expected value (or mean) of a discrete random variable:

$$\mu = E(X) = \sum_i x_i f(x_i)$$

Interpretation:

- Weighted average of all possible values
- Each value weighted by its Probability Distribution
- May not be an actual possible value of X

Example: Expected Value Calculation

Given:

x	0	1	2	3	4
$f(x)$	1/5	1/5	1/5	1/5	1/5

Step-by-step calculation:

$$\begin{aligned}
 \mu = E(X) &= \sum x f(x) \\
 &= 0 \left(\frac{1}{5} \right) + 1 \left(\frac{1}{5} \right) + 2 \left(\frac{1}{5} \right) + 3 \left(\frac{1}{5} \right) + 4 \left(\frac{1}{5} \right) \\
 &= \frac{0 + 1 + 2 + 3 + 4}{5} \\
 &= \frac{10}{5} = 2
 \end{aligned}$$

Note: In this case, expected value equals a possible value of X (*visualization*).

Variance and Standard Deviation

Variance of Discrete Random Variable

$$\sigma^2 = \text{Var}(X) = E[(X - E[X])^2] = \sum_i (x_i - \mu)^2 f(x_i)$$

Computational formula (easier):

$$\sigma^2 = \text{Var}(X) = \sum_i x_i^2 f(x_i) - \mu^2$$

Standard Deviation

$$\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}$$

Note: If $0 < \sigma^2 < 1$, then $\sigma > \sigma^2$

Example: Variance and Standard Deviation

Given: Same example with $\mu = 2$

Step 1: Calculate $\sum x_i^2 f(x_i)$

$$\begin{aligned}\sum x^2 f(x) &= 0^2 \left(\frac{1}{5}\right) + 1^2 \left(\frac{1}{5}\right) + 2^2 \left(\frac{1}{5}\right) + 3^2 \left(\frac{1}{5}\right) + 4^2 \left(\frac{1}{5}\right) \\ &= \frac{0 + 1 + 4 + 9 + 16}{5} = \frac{30}{5} = 6\end{aligned}$$

Step 2: Apply formula

$$\begin{aligned}\sigma^2 &= \sum x^2 f(x) - \mu^2 = 6 - 2^2 = 6 - 4 = 2 \\ \sigma &= \sqrt{2} \approx 1.414\end{aligned}$$

Exercise: Prior Convictions

Context: Study of 500 prisoners at a state prison

X = number of prior convictions ($x = 0, 1, 2, 3, 4$)

$X = x$	0	1	2	3	4
Number of Prisoners	80	265	100	40	15
$f(x) = P(X = x)$	80/500	265/500	100/500	40/500	15/500
$f(x)$ (decimal)	0.16	0.53	0.20	0.08	0.03

Questions:

- 1 What is the expected number of prior convictions?
- 2 What is the standard deviation?
- 3 What is $P(X \leq 2)$?

Binary Variables

Binary Categorical Variable

A variable that has exactly **two possible outcomes**

Examples:

- Pass/Fail on an exam
- Defective/Non-defective part
- Success/Failure of a system test
- Yes/No response to a survey

Key Insight:

- We define one outcome as "success" and the other as "failure"
- These labels are arbitrary (researcher's choice)
- "Success" doesn't necessarily mean positive outcome!

The Binomial Experiment

A binomial experiment requires ALL four conditions:

- ➊ **Fixed number of trials:** n is predetermined and constant
- ➋ **Two outcomes only:** Each trial results in "success" or "failure"
- ➌ **Constant Probability Distribution:** Probability Distribution of success p is same for all trials
- ➍ **Independence:** Outcome of one trial doesn't affect others

Important

If ANY condition is violated, it is NOT a binomial experiment!

Binomial Random Variable

Definition

If all four binomial conditions are satisfied, then:

X = number of successes in n trials

is a binomial random variable

Parameters:

- n = number of trials (fixed)
- p = Probability Distribution of success on each trial
- $q = 1 - p$ = Probability Distribution of failure

Notation: $X \sim \text{Binomial}(n, p)$ or $X \sim B(n, p)$

Binomial Probability Distribution Formula

PMF of Binomial Distribution

For $X \sim B(n, p)$:

$$f(x) = P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x} \quad \text{for } x = 0, 1, 2, \dots, n$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Components:

- $\binom{n}{x}$ = number of ways to arrange x successes in n trials
- p^x = Probability Distribution of x successes
- $(1 - p)^{n-x}$ = Probability Distribution of $n - x$ failures

Recall: $n! = n \times (n - 1) \times \dots \times 2 \times 1$, and $0! = 1$, $1! = 1$

Binomial Distribution: Key Formulas

For binomial random variable $X \sim B(n, p)$:

Mean (Expected Value)

$$\mu = E(X) = np$$

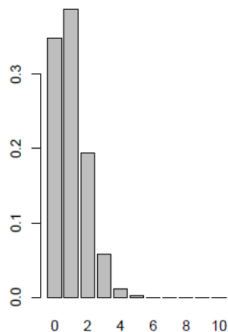
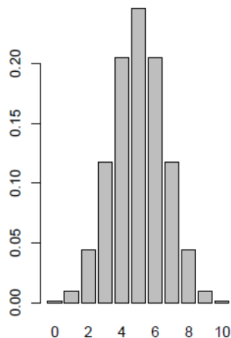
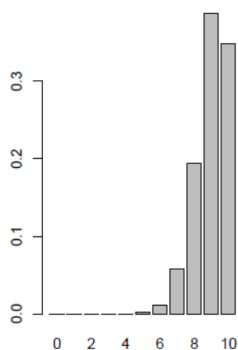
Variance

$$\sigma^2 = \text{Var}(X) = np(1 - p) = npq$$

Standard Deviation

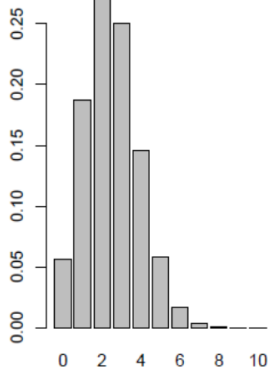
$$\sigma = \text{SD}(X) = \sqrt{np(1 - p)} = \sqrt{npq}$$

Graphical Displays of Binomial Distributions

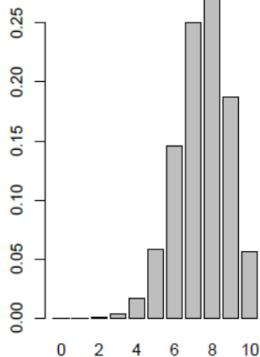
Binomial $n = 10$ $p = 0.1$ Binomial $n = 10$ $p = 0.5$ Binomial $n = 10$ $p = 0.9$ 

Graphical Displays of Binomial Distributions

Binomial $n=10$ $p=0.25$ [link](#)



Binomial $n=10$ $p=0.75$



Example: Crime Survey Setup

Context: An FBI survey shows that about 80% of all property crimes go unsolved. Suppose that in your town 3 such crimes are committed and they are each deemed independent of each other. What is the probability that 1 of 3 of these crimes will be solved?

Check binomial conditions:

- 1 **Fixed n ?** Yes, $n = 3$ crimes
- 2 **Two outcomes?** Yes, solved or unsolved
- 3 **Constant p ?** Yes, $p = 0.2$ (Probability Distribution of being solved)
- 4 **Independent?** Yes, stated in problem

Conclusion

This IS a binomial experiment with $n = 3$ and $p = 0.2$

Define: X = number of crimes solved, where $X \sim B(3, 0.2)$

Example: Exactly 1 Crime Solved

Find: $P(X = 1)$ where $X \sim B(3, 0.2)$

Method 1: Enumerate sequences

Sequence 1: S, U, U = $(0.2)(0.8)(0.8) = 0.128$

Sequence 2: U, S, U = $(0.8)(0.2)(0.8) = 0.128$

Sequence 3: U, U, S = $(0.8)(0.8)(0.2) = 0.128$

Total: $P(X = 1) = 3(0.128) = 0.384$

Method 2: Use formula

$$\begin{aligned}P(X = 1) &= \binom{3}{1} (0.2)^1 (0.8)^2 \\&= \frac{3!}{1!2!} (0.2)(0.64) \\&= 3 \times 0.128 = 0.384\end{aligned}$$

Example: At Least One Solved

Find: $P(X \geq 1)$ where $X \sim B(3, 0.2)$

Method 1: Direct (long way)

$$\begin{aligned}P(X \geq 1) &= P(X = 1) + P(X = 2) + P(X = 3) \\&= 0.384 + 0.096 + 0.008 = 0.488\end{aligned}$$

Method 2: Complement (short way)

$$\begin{aligned}P(X \geq 1) &= 1 - P(X < 1) = 1 - P(X = 0) \\&= 1 - \binom{3}{0} (0.2)^0 (0.8)^3 \\&= 1 - 1(1)(0.512) \\&= 1 - 0.512 = 0.488\end{aligned}$$

Tip: Using complements is often much easier!

Crime Survey: Mean and Standard Deviation

For unsolved crimes: $n = 3$, $p = 0.8$ (Probability Distribution unsolved)

Expected value:

$$\mu = E(X) = np = 3(0.8) = 2.4$$

Interpretation: On average, 2.4 of the 3 crimes remain unsolved

Variance and standard deviation:

$$\begin{aligned}\sigma^2 &= np(1 - p) = 3(0.8)(0.2) = 0.48 \\ \sigma &= \sqrt{0.48} \approx 0.693\end{aligned}$$

Note: Expected value may not be a possible integer value of X

Binomial Distribution: Shape

The shape depends on n and p :

- When $p = 0.5$: Distribution is **symmetric**
- When $p < 0.5$: Distribution is **right-skewed**
- When $p > 0.5$: Distribution is **left-skewed**
- As n increases: Distribution becomes more bell-shaped

Key Insight for Engineering:

- If testing components with low failure rate (p small), distribution will be highly right-skewed
- If testing components with high reliability (p large for success), distribution will be left-skewed
- Larger sample sizes yield more normal-looking distributions

The Normal Distribution

Normal Distribution

A family of continuous distributions that model many real-world phenomena. Characterized by:

- Bell-shaped
- Symmetric about the mean

Two parameters:

- 1 μ = mean (center of curve)
- 2 σ^2 = variance (σ = standard deviation; controls spread)

Notation: $X \sim N(\mu, \sigma^2)$

Properties:

- Range: $-\infty$ to $+\infty$
- Total area under curve = 1

Standard Normal Distribution

Standard Normal Distribution

A special normal distribution with:

- Mean: $\mu = 0$
- Variance: $\sigma^2 = 1$ (thus $\sigma = 1$)

Notation: $Z \sim N(0, 1)$

Why is it important?

- **Any** normal distribution can be transformed to standard normal
- Allows use of standard tables for all normal calculations
- Simplifies Probability Distribution computations

Z-score transformation:

$$Z = \frac{X - \mu}{\sigma}$$

Z-scores: Definition and Interpretation

Z-score

The Z-score represents the **number of standard deviations** an observation is from the mean:

$$Z = \frac{X - \mu}{\sigma}$$

Example: NMEC 595 final score, $\mu = 68.55$, $\sigma = 15.45$

- Score 80: $Z = \frac{80-68.55}{15.45} = 0.74$
 $\Rightarrow$ 0.74 SD **above** mean
- Score 60: $Z = \frac{60-68.55}{15.45} = -0.55$
 $\Rightarrow$ 0.55 SD **below** mean

Characteristics:

- Positive Z : above mean
- Negative Z : below mean
- $Z > 3$ or $Z < -3$: potential outliers

Using Standard Normal Table

Standard Normal Tables provide: $P(Z \leq z)$ (left-tail probabilities)

Example: Find $P(Z \leq 0.87)$

Steps:

- 1 Locate row for first two digits: 0.8
- 2 Locate column for second decimal: 0.07
- 3 Read value at intersection: 0.8078

Therefore: $P(Z \leq 0.87) = 0.8078$

Interpretation: 80.78% of values fall below $z = 0.87$

Software alternative: Minitab, R, Python can compute these directly

Standard Normal: Greater Than

Find: $P(Z > 0.87)$

Method: Use complement rule

$$\begin{aligned}P(Z > 0.87) &= 1 - P(Z \leq 0.87) \\&= 1 - 0.8078 \\&= 0.1922\end{aligned}$$

Interpretation: 19.22% of values fall above $z = 0.87$

Key Principle: Tables give "less than" probabilities. Use complements for "greater than."

Standard Normal: Between Two Values

Find: $P(2 < Z < 3)$

Method: Subtract CDFs

$$\begin{aligned}P(2 < Z < 3) &= P(Z < 3) - P(Z \leq 2) \\&= 0.9987 - 0.9772 \\&= 0.0215\end{aligned}$$

Interpretation: 2.15% of values fall between $z = 2$ and $z = 3$

General formula:

$$P(a < Z < b) = P(Z < b) - P(Z < a) = F(b) - F(a)$$

Finding Percentiles

Question: Find the 10th percentile of standard normal

Meaning: Find z^* such that $P(Z \leq z^*) = 0.10$

Method:

- 1 Search table body for value closest to 0.10
- 2 Closest value: 0.1003
- 3 Read margins: row = -1.2, column = 0.08
- 4 Therefore: $z^* = -1.28$

Verification: $P(Z \leq -1.28) = 0.1003 \approx 0.10$

Interpretation: 10% of standard normal values fall below $z = -1.28$

General Normal to Standard Normal

Problem: Given $X \sim N(\mu, \sigma^2)$, find probabilities

Solution: Transform to standard normal using Z-score

Standardization Formula

$$Z = \frac{X - \mu}{\sigma}$$

where $Z \sim N(0, 1)$

Then:

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right)$$

Process:

- 1 Convert x values to z scores
- 2 Use standard normal table/software
- 3 Interpret result

Example: Female Heights

Given: U.S. adult female heights $\sim N(64, 2^2)$ inches

Find: Probability Distribution a randomly selected female is shorter than 65 inches

Step 1: Compute Z-score

$$Z = \frac{X - \mu}{\sigma} = \frac{65 - 64}{2} = \frac{1}{2} = 0.5$$

Step 2: Find Probability Distribution

$$P(X < 65) = P(Z < 0.5) = 0.6915$$

Interpretation: About 69.15% of U.S. adult females are shorter than 65 inches

Example: Weight Between Two Values

Given: 10-year-old girls' weights $\sim N(70, 13^2)$ pounds

Find: $P(60 < X < 90)$

Step 1: Convert to Z-scores

$$\text{For } x = 60 : \quad z = \frac{60 - 70}{13} = -0.77$$

$$\text{For } x = 90 : \quad z = \frac{90 - 70}{13} = 1.54$$

Step 2: Find Probability Distribution

$$\begin{aligned} P(60 < X < 90) &= P(-0.77 < Z < 1.54) \\ &= P(Z < 1.54) - P(Z < -0.77) \\ &= 0.9382 - 0.2206 = 0.7176 \end{aligned}$$

Interpretation: 71.76% of 10-year-old girls weigh between 60 and 90 pounds

Finding Percentiles: General Normal

Problem: Find 60th percentile of $X \sim N(70, 13^2)$

Step 1: Find 60th percentile of standard normal

- Find z^* such that $P(Z \leq z^*) = 0.60$
- From table: $z^* \approx 0.25$

Step 2: Back-transform to original scale

$$z = \frac{x - \mu}{\sigma}$$
$$0.25 = \frac{x - 70}{13}$$
$$x = 70 + 0.25(13) = 70 + 3.25 = 73.25$$

Answer: 60th percentile is 73.25 pounds

Interpretation: 60% of 10-year-old girls weigh 73.25 pounds or less

The Empirical Rule (68-95-99.7)

Empirical Rule

For any normal distribution, approximately:

- **68%** of values fall within $\mu \pm 1\sigma$
- **95%** of values fall within $\mu \pm 2\sigma$
- **99.7%** of values fall within $\mu \pm 3\sigma$

Visual interpretation:

- About 2/3 of data within 1 SD of mean
- Almost all data (95%) within 2 SD of mean
- Nearly all data (99.7%) within 3 SD of mean

Usage: Quick approximations; use tables/software for precision

Empirical Rule: Example

Given: Component lifetimes $\sim N(1000, 100^2)$ hours

Apply Empirical Rule:

- $\mu = 1000, \sigma = 100$
- $\mu \pm 1\sigma = 1000 \pm 100 = [900, 1100]$ hours $\rightarrow 68\%$
- $\mu \pm 2\sigma = 1000 \pm 200 = [800, 1200]$ hours $\rightarrow 95\%$
- $\mu \pm 3\sigma = 1000 \pm 300 = [700, 1300]$ hours $\rightarrow 99.7\%$

Interpretations:

- About 68% of components last between 900-1100 hours
- About 95% last between 800-1200 hours
- Almost all (99.7%) last between 700-1300 hours
- Component lasting < 700 or > 1300 hours is very unusual

Verifying Empirical Rule

Check rule using standard normal table:

For 68%:

$$\begin{aligned}P(-1 < Z < 1) &= P(Z < 1) - P(Z < -1) \\&= 0.8413 - 0.1587 = 0.6826 \approx 0.68\end{aligned}$$

For 95%:

$$\begin{aligned}P(-2 < Z < 2) &= P(Z < 2) - P(Z < -2) \\&= 0.9772 - 0.0228 = 0.9544 \approx 0.95\end{aligned}$$

For 99.7%:

$$\begin{aligned}P(-3 < Z < 3) &= P(Z < 3) - P(Z < -3) \\&= 0.9987 - 0.0013 = 0.9974 \approx 0.997\end{aligned}$$

Summary: Discrete vs Continuous

Aspect	Discrete	Continuous
Values	Countable	Uncountable (interval)
Probability Distribution function $P(X = x)$	PMF: $f(x) = P(X = x)$ Can be > 0	PDF: $f(x) \neq P(X = x)$ Always 0
Cumulative	$F(x) = \sum P(X \leq x)$	$F(x)$ = area under PDF
Examples	Binomial, Poisson	Normal, t, χ^2 , F

Common thread:

- Both have mean $\mu = E(X)$ and variance σ^2
- Both use CDF: $F(x) = P(X \leq x)$
- Both fundamental for statistical inference

Key Formulas: Quick Reference

Discrete Random Variable:

$$\mu = E(X) = \sum x f(x)$$

$$\sigma^2 = \sum x^2 f(x) - \mu^2$$

Binomial Distribution: $X \sim B(n, p)$

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

$$\mu = np, \quad \sigma^2 = np(1 - p)$$

Normal Distribution: $X \sim N(\mu, \sigma^2)$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$P(X \leq x) = P\left(Z \leq \frac{x - \mu}{\sigma}\right)$$

Important Concepts to Remember

- 1 **Random variables** quantify random outcomes numerically
- 2 **Binomial** requires: fixed n , two outcomes, constant p , independence
- 3 **Expected value** is weighted average (may not be possible value)
- 4 **Standard deviation** measures typical spread from mean
- 5 **Normal distribution** is ubiquitous (Central Limit Theorem)
- 6 **Z-scores** standardize any normal to $N(0, 1)$
- 7 **Empirical Rule:** 68-95-99.7 for quick approximations
- 8 **Tables and software** are essential tools for calculations

Applications in Mechanical Engineering Research

Where these distributions appear:

- **Binomial:** Pass/fail testing, quality control, reliability studies
- **Normal:** Measurement errors, dimensional tolerances, material properties

Next steps:

- Use these foundations for hypothesis testing
- Develop confidence intervals
- Apply to regression and experimental design